Product Requirements Document - Project Falcon
Company: Acme Robotics, Inc.
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Status: Approved

1. Overview

Project Falcon is Acme Robotics' second-generation autonomous home cleaning robot, codenamed "Falcon." It succeeds the "Sparrow" model launched in 2024. Falcon is designed for medium to large homes up to 2,500 square feet and introduces LiDAR-based mapping, a self-emptying dock, and a mobile app with room-specific scheduling.

2. Goals and Success Metrics

- Achieve a battery life of at least 150 minutes per charge on the Standard cleaning mode.
- Reduce customer support tickets related to navigation errors by 40 percent compared to Sparrow.
- Reach 100,000 units sold within the first six months of launch.
- Maintain a customer satisfaction (CSAT) score of 4.5 out of 5 or higher.

3. Target Launch Date

Falcon is scheduled for general availability on September 15, 2026, with a limited early-access rollout beginning August 1, 2026 for 5,000 waitlisted customers.

4. Team

- Product Manager: Maria Chen
- Engineering Lead: Devon Okafor
- Hardware Lead: Priya Raman
- QA Lead: Tom Whitfield

The Falcon team consists of 14 engineers across firmware, mobile, and mechanical design.

5. Budget

The total approved budget for Project Falcon is $8.2 million, allocated as follows: $4.5 million for hardware R&D, $2.1 million for software development, $1.0 million for marketing, and $0.6 million for contingency.

6. Key Features

- LiDAR navigation with room mapping, up to 5 floors saved per household
- Self-emptying dock with a 60-day dust bag capacity
- Battery life: 150 minutes on Standard mode, 90 minutes on Max Power mode
- Suction power: 4,000 Pa
- Weight: 3.8 kg
- Retail price: $649
- Companion mobile app supporting iOS 16+ and Android 12+
- Voice control via Alexa and Google Assistant integration

7. Out of Scope for v1

- Multi-robot coordination, planned for Falcon v2, targeted for 2027
- Outdoor or patio cleaning mode

- Integration with third-party smart-home hubs other than Alexa and Google Assistant

8. Risks

- The LiDAR sensor supplier, OptiSense Corp, has a lead time of 16 weeks, which could delay hardware production if orders are not placed by April 30, 2026.
- An industry-wide battery cell shortage could affect the Max Power mode battery target.

9. Approval

This document was approved by the Product Steering Committee on March 20, 2026.